Fpt University

TRƯỜNG ĐẠI HỌC FPT

CAPSTONE PROJECT REPORT

Report 2 – Project Management Plan

– Ho Chi Minh City, October 2026 –

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I. Record of Changes

Date	A* M, D	In charge	Change Description
28/10/2026	A	Lê Thành Minh Trung	First version for Review 2: scope and estimation, objectives, risks, process, quality, training, deliverables, responsibilities, communication, configuration management
29/10/2026	A	Võ Minh Luân	Added Responsibility Assignments and Project Communications sections.
29/10/2026	M	Trần Võ Quang Duy	Tools & Infrastructures: updated the mobile technology to Flutter.
29/10/2026	M	Nguyễn Xuân Lộc	Training Plan: Android training changed from Kotlin and Jetpack Compose to Flutter and Dart.

*A - Added M - Modified D - Deleted

II. Project Management Plan

1. Overview

1.1 Scope & Estimation

The scope is the registration form version 2.0 (Record of Changes BM03) and the functional requirements of Report 3. Complexity: *Simple* = one screen or endpoint with plain validation (2 man-days); *Medium* = several screens or rules, one state machine (4 man-days); *Complex* = cross-feature rules, jobs, money or the assignment engine (7–10 man-days). Effort includes design, code, automated tests and review.

#	WBS Item	Complexity	Est. Effort (man-days)
1	**Project management, requirements and documents**		**40**
1.1	Planning, tracking, meetings, reviews	Medium	12
1.2	Requirements (Brief, Change Specification v2), Reports 1–5, Review 2 forms	Complex	18
1.3	Reports 6–7, defense slides	Medium	10
2	**Accounts and access (AUTH)**		**27**
2.1	Sign-in with password and Google, sessions, lockout, password reset	Complex	7
2.2	Student verification, student list import (CSV, XLSX)	Complex	7
2.3	Users, roles, permission matrix, Staff accounts, Audit Log	Medium	6
2.4	Enrollment reconciliation	Complex	7
3	**Registration periods and configuration (CFG)**		**25**
3.1	Room Types, Room Classes, fees, Utility, parameters	Medium	4
3.2	Rules: eligibility, priority, matching, deadlines, refunds, stay	Complex	7
3.3	Registration-period requests	Medium	4
3.4	Term wizard and publication, change announcements	Complex	10
4	**Inventory (INV)**		**19**
4.1	Blocks, Floors, Rooms, Beds, out of service, bulk generation	Medium	4
4.2	Room Class per Room per Term	Medium	4
4.3	Furniture types, items, moves, QR labels	Medium	4
4.4	Door codes (encrypted, rotation, release)	Complex	7
5	**Applications (APP)**		**19**
5.1	Eligibility and the Application wizard with the Deposit Invoice	Complex	7
5.2	Roommate invitations	Medium	4
5.3	Review, cancellation, expiry	Medium	4
5.4	Round 2	Medium	4
6	**Assignment and residence (ASG)**		**36**
6.1	Assignment engine, runs, conflicts, approval	Complex	10
6.2	Balance, forfeit, residence, door-code release	Complex	7

#	WBS Item	Complexity	Est. Effort (man-days)
6.3	Check-in and check-out desk, settlement, damage per item	Complex	7
6.4	Late check-in, terminations	Medium	4
6.5	Operational moves, turnover	Complex	8
7	**Payments and billing (BIL)**		**25**
7.1	Invoices, VietQR payment page, payment claims and confirmation	Complex	8
7.2	Meter readings, Utility Invoices split by days	Complex	7
7.3	Overdue, reminders, Utility debt warnings, refunds	Complex	10
8	**Term transition and requests (ASG, REQ)**		**30**
8.1	Requests window M2–M7, next-Term Allocations, Stays	Complex	10
8.2	Room change, swap and swap board, Room Class change, extension	Complex	10
8.3	Next-Term fees; routing rules, time limits, escalations	Complex	10
9	**Maintenance and inspections (MNT)**		**11**
9.1	Maintenance Requests with photos and furniture items	Medium	4
9.2	Inspection checklists, schedules and runs	Complex	7
10	**Notifications (NTF)**		**15**
10.1	In-app notifications, templates in two languages, email outbox	Medium	4
10.2	FCM push, device tokens	Medium	4
10.3	Announcements and the dormitory board, email campaigns	Complex	7
11	**Reports (RPT)**		**12**
11.1	Dashboards per role, Term dashboard	Complex	7
11.2	Reports, CSV exports, search	Medium	5
12	**Clients**		**42**
12.1	Web design system, layouts, menus, two languages	Complex	10
12.2	Web screens of the four roles (88 screens)	Complex	20
12.3	Android app for students	Complex	12
13	**Infrastructure and quality**		**35**
13.1	Architecture, database design and migrations	Complex	8
13.2	CI/CD, Docker, Vercel and Azure environments, runbook, demo data tool	Complex	10
13.3	System tests, NFR measurements, end-to-end automation	Complex	17
	Total Estimated Effort (man-days)		**336**

1.2 Project Objectives

Overall objective. Deliver CampusStay FPT, a dormitory system for FPT University's Tân Uyên campus that covers registration, assignment, payments, residence, Term transitions, maintenance, Utility billing and reports for four roles, on the web and Android, deployed and demonstrable, with the

documents of the capstone, by the final defense; and pass Defense 1.1 with every committed workflow, test and NFR target of the Review 2 Commitment.

Quality.

#	Testing Stage	Test Coverage	No. of Defects	% of Defect	Notes
1	Reviewing	100 % of changes reviewed before `main`; self-review prompt on each review version	≤ 15	30 %	Design, document and code review; BM02 findings
2	Unit Test	Pure logic classes (engine, split, exchanges, exports)	≤ 5	10 %	`mvn test` on every change
3	Integration Test	100 % of endpoints; every FR with ≥ 1 test; ≥ 90 % pass (target 100 %)	≤ 20	40 %	`mvn verify` in CI; failing build blocks deploy
4	System Test	100 % of main workflows; 100 % main and ≥ 80 % exception demo scenarios	≤ 8	15 %	Staging with demo data
5	Acceptance Test	Demo scenarios with the mentor and the council	≤ 2	5 %	No open Critical or High defect

Milestone timeliness: ≥ 90 % of milestones of section 3 delivered on their date; review documents delivered at least one day before the review.

Allocated effort (man-days, % of 336): project management 26 (8 %), requirements 34 (10 %), design 47 (14 %), coding 162 (48 %), testing 50 (15 %), deployment and documentation of operations 17 (5 %).

1.3 Project Risks

#	Risk Description	Impact	Possibility	Response Plans
1	The dormitory is not operating yet: the real rules may differ from the requirements	High	Medium	Requirements from FPT's operating dormitories, Circular 27/2011 and VNU-HCM practice; every business value configurable; assumptions listed in Report 3; weekly mentor review
2	Most of Iteration 1 was written by the leader: other members may not master their parts (bus factor)	High	High	Ownership map (section 4); each member owns 2–3 parts and one workflow, takes the training plan (2.3), explains and changes their part before Defense 1.1
3	Manual confirmation of payment claims at the registration peak	Medium	Medium	Claims stop the deadline; bulk confirmation with count and total; hourly summary
4	One small Azure VM and a Burstable database under load	Medium	Medium	NFR-01 load test, health check and rollback; larger VM documented
5	Azure for Students credit or free services end before the final defense	High	Low	Budget alert at $5/month; runbook to move to another VPS with Docker Compose in a few hours
6	Third-party services change or fail (Google sign-in, FCM, Gmail)	Medium	Low	Each behind a port with a configuration switch; the system works without push or email; tests with stubs
7	Schedule: defense dates not announced, end-to-end tests	Medium	Medium	Sprints S5–S6 reserved for them (section 3); tracked in Project Tracking

#	Risk Description	Impact	Possibility	Response Plans
	and NFR measurements still to do			
8	Personal data protection (Law 91/2025/QH15)	High	Low	Fictional data only in seeds, tests and the demo; access by role; no personal data in URLs or logs; demo emails redirected

2. Management Approach

The team works in two-week sprints (Scrum-like, 8 sprints from 07/09/2026), with the backlog in the Project Tracking workbook and GitHub. The leader acts as product owner and Scrum master; the mentor reviews the increment every week.

2.1 Project Process

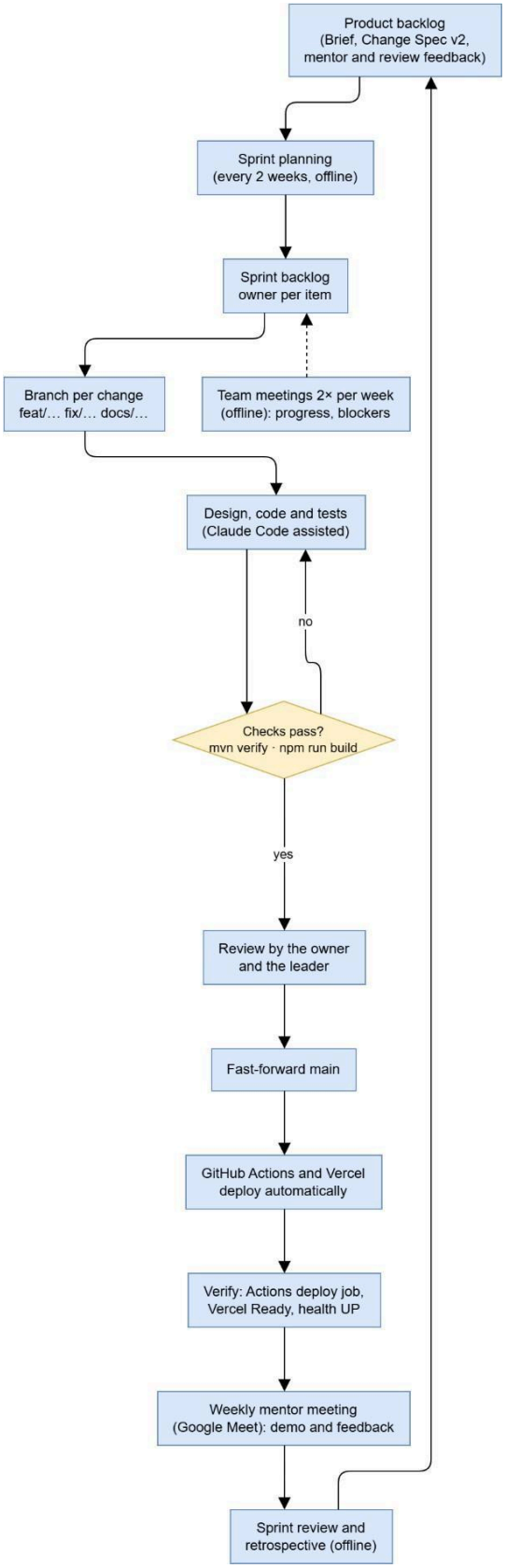


Figure 1: Development process

1. **Backlog.** Items come from the registration form, the Brief and the Change Specification v2, the mentor's weekly feedback and the reviews (Review 1 added the Manager role and the requirement basis). Each item has an owner (section 4).
2. **Sprint planning** at the start of each sprint (offline): items, owners, acceptance criteria and test cases.
3. **Development** on a branch per change (`feat/…`, `fix/…`, `docs/…`); until Review 2 each phase of the Change Specification v2 had its own branch `feat/v2-phase<n>`. Code, tests and documentation change together.
4. **Checks**: `mvn -o verify` for any backend change, `npm run build` for any web change, Gradle build for Android. A failing check stops the change.
5. **Review and merge**: the owner and the leader review; `main` is fast-forwarded; GitHub Actions deploys the API and Vercel the web automatically; the result is verified (deploy job, Vercel Ready, health UP).
6. **Mentor meeting** every week on Google Meet: demo of the increment, questions (Q&A sheet), feedback into the backlog.
7. **Team meetings** twice a week (offline): progress, blockers, pairing on parts a member is learning. Sprint review and retrospective at the end of each sprint.

2.2 Quality Management

- **Defect prevention.** Decisions recorded as ADRs and in the Change Specification before coding; business rules as configuration; database constraints for the critical rules; layered architecture checked by `LayeringTest`; one status palette and one message catalogue in two languages.
- **Reviewing.** Every change is reviewed before `main`; documents are generated from the code where possible (screen map, endpoint map, test catalog, ERDs) so they cannot drift; before each review the standard self-review prompt is run on the tagged version and every finding gets a response (BM02).
- **Unit testing.** Pure logic (assignment engine, Utility split, Google code exchange, FCM sender, exports, message translation) tested without a database.
- **Integration testing.** Every endpoint through HTTP with a real PostgreSQL (Testcontainers); permissions and scope for each role; jobs driven by a test clock; the whole suite in CI on every push.
- **System testing.** The five main workflows end to end on staging with the demo stages; NFR measurements; scenario-based demo rehearsal before each defense.

2.3 Training Plan

Training Area	Participants	When, Duration	Waiver Criteria
Domain: dormitory operations, the Change Specification v2, the five workflows	All (mandatory)	29/10 – 05/11/2026, 2 sessions of 2 h, led by the leader	None
Java 21, Spring Boot 4, JdbcClient, Flyway, the layered packages	Nguyễn Xuân Lộc, Võ Minh Luân (mandatory)	02/11 – 15/11/2026, 4 sessions of 2 h + one change in their part	Has merged a reviewed backend change with tests
React 19, Tailwind 4, the web design system and i18n	Trần Võ Quang Duy (mandatory), others optional	02/11 – 15/11/2026, 3 sessions of 2 h	Has merged a reviewed web change

Training Area	Participants	When, Duration	Waiver Criteria
Kotlin, Jetpack Compose, FCM	Võ Minh Luân (mandatory)	02/11 – 15/11/2026, 3 sessions of 2 h	Has built and run the Android app with push
Testing: JUnit 5, Testcontainers, writing a test case from an acceptance criterion	All (mandatory)	09/11 – 20/11/2026, 2 sessions of 2 h	Owns ≥ 3 passing automated tests
Git and GitHub flow, commit format, CI/CD and deployment checks	All (mandatory)	29/10/2026, 1 session of 2 h	Has pushed a branch that passed CI
AI tools (Claude Code): rules of use, review of AI output, no personal data or secrets	All (mandatory)	29/10/2026, 1 h	None

3. Project Deliverables

#	Deliverable	Due Date	Notes
1	Registration form version 1.0	28/08/2026	Approved topic FA26SE260
2	Review 1 (presentation, no reports)	30/09/2026	Feedback: requirement basis, Manager role
3	Iteration 1: API, web app, production deployment, Change Specification v2 phases 1–8	27/10/2026	Tag `review-2`
4	Review 2 documents: Registration form 2.0 + BM03, Reports 1–5, Project Tracking, BM01 Design Brief, BM02 Self-Review	30/10/2026	PDF, `SWP490_FA26SE260_R2_<name>.pdf`
5	Review 2	31/10/2026	
6	BM04 Review 2 Commitment	05/11/2026	Within 3 working days after the review
7	Iteration 2: end-to-end workflow tests, NFR measurements, concurrency test, Android tests, fixes from Review 2	29/11/2026	Sprints S5–S6
8	Report 6 (User Guides), updated Reports 1–5, Full Software Package	13/12/2026	Before Defense 1.1 (date to be announced)
9	Defense 1.1	TBD	Thresholds of the Review 2 Commitment
10	Report 7 (Final Report), slides, final package; final defense (Viva)	TBD (duration ends 03/2027)	

4. Responsibility Assignments

D = Do; R = Review; S = Support; I = Informed; blank = omitted.

Responsibility	Lê Thành Minh Trung (SE183752)	Nguyễn Xuân Lộc (SE183782)	Trần Võ Quang Duy (SE182146)	Võ Minh Luân (SE182794)
Project planning and tracking	D	S	S	S

Report 1 Project Introduction	D	R	S	S
Report 2 Project Management Plan	D	R	R	R

Responsibility	Lê Thành Minh Trung (SE183752)	Nguyễn Xuân Lộc (SE183782)	Trần Võ Quang Duy (SE182146)	Võ Minh Luân (SE182794)
Report 3 SRS (overview, use cases, business rules)	D	S	S	R
Report 3 SRS (screens, authorization, messages)	R	S	D	S
Report 4 SDD (architecture, database, DevOps)	D	S	I	I
Report 4 SDD (state machines, activity diagrams, detailed design)	D	S	S	S
Report 5 Test Documentation, Test Report	R	S	S	D
Architecture, database, CI/CD and deployment	D	S	I	S
Applications, payments, assignment, residence (APP, BIL, ASG)	D	S	S	I
Term transition, next-Term requests, reconciliation	R	D	I	S
Registration periods, Term wizard, inventory screens, web UI and i18n	R	S	D	I
Android app, push, maintenance and escalations	R	S	I	D
System tests, demo scenarios, NFR measurements	R	S	S	D
Review 2 forms (BM01–BM04), self-review	D	R	R	R

5. Project Communications

Communication Item	Who / Target	Purpose	When, Frequency	Type, Tool, Method
Team meeting	All members	Progress, blockers, pairing, decisions	Twice a week, 1–2 h	Offline at campus
Mentor meeting	Team, mentor Thân Thị Ngọc Vân	Demo of the increment, questions, feedback	Weekly, 1 h	Online, Google Meet; minutes in Q&A and Issues sheets
Sprint planning, review and retrospective	All members	Plan the sprint, review the increment, improve the process	Every 2 weeks	Offline; backlog in Project Tracking

Daily status	All members	Short updates, questions	Daily, asynchronous	Team chat group

Communication Item	Who / Target	Purpose	When, Frequency	Type, Tool, Method
Change review	Owner, leader	Review of code and documents before `main`	Each change	GitHub branches and commits; CI results
Reviews and defenses	Team, reviewers, council	Approval of scope, design and results	Review 2: 31/10/2026; others announced	Offline; documents in PDF

6. Configuration Management

6.1 Document Management

- All project documents live in the repository under `docs/`: the Brief, the Change Specification v2, the API contract, the backend and web design notes, the report sources (`docs/reports/src/`*`.md) and the diagrams as draw.io files (`docs/diagrams/`*`.drawio`).
- Report PDFs are generated from these sources into the school's Word templates (`node docs/reports/src/build.mjs`), so every version is reproducible; tables that must match the code (screens, messages, test cases, ERDs) are generated from the code.
- Each document has a Record of Changes; review versions are frozen by the Git tag of the review (`review-2`); file names follow `SWP490_FA26SE260_R2_<name>.pdf`.

6.2 Source Code Management

- One GitHub monorepo `tisou47/CampusStay-FPT` (`backend/`, `web/`, `android/`, `docs/`, `infra/`, `.github/workflows/`).
- `main` is production; work is done on branches (`feat/…`, `fix/…`, `docs/…`); `main` is fast-forwarded only after the checks pass; commits follow `type(feature): …` (`feat(web): …`, `fix(bil): …`).
- Database changes only through new Flyway migrations; secrets never in Git (`.env.example` lists the settings); tags mark review versions.

6.3 Tools & Infrastructures

Category	Tools / Infrastructure
Technology	Java 21, Spring Boot 4.1, PostgreSQL 17, React 19, Vite 8, Tailwind CSS 4, Kotlin 2.1, Jetpack Compose
IDEs and editors	IntelliJ IDEA, VS Code, Android Studio
AI tools	Claude Code (Anthropic) — specification, design, code, tests, review (Report 4 section 1.3.5)
Design	Figma (file *CampusStay-FPT*), draw.io (UML diagrams, ERDs in Chen's notation)
Source control and CI/CD	Git, GitHub, GitHub Actions, GitHub Container Registry
Hosting	Vercel (web), Microsoft Azure for Students (VM with Docker Compose and Caddy, Azure Database for PostgreSQL)

Testing	JUnit 5, AssertJ, Testcontainers, Postman
Project management and documents	Project Tracking workbook, Microsoft Word and Excel (school templates), Google Meet